

A Hybrid Neuro-Symbolic Reasoning Framework for Automated Factual Verification in Conversational AI

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Abstract:

This paper presents a novel framework for automated factual verification in conversational AI systems through the integration of symbolic logic, statistical semantics, and graph-based reasoning. Unlike conventional approaches that rely solely on either rule-based systems or large language models (LLMs), our hybrid methodology combines the formal guarantees of logical verification with the flexibility of neural language understanding. We implement this framework as a comprehensive reasoning engine that can ingest organizational policies as documents and automatically validate LLM-generated responses against these policies. The system exhibits improved alignment with organizational policies through its ability to handle natural language variations, perform numeric threshold validations, and execute multi-hop reasoning over complex rules. We discuss the implications of this work for developing more reliable, trustworthy AI assistants that maintain factual consistency with authoritative knowledge sources while preserving the natural conversational capabilities of modern LLMs.

Lewis et al. [13] conclude that parametric-only models like BART generate more factual and

specific language than retrieval-augmented approaches on knowledge-intensive generation tasks. The policy optimization in Stiennon et al. [9] was performed using the A2C reinforcement learning algorithm, which allowed for stable training across different model scales. [8] Model cards are designed to disclose the context in which models are intended to be used, details of the performance evaluation procedures, and other relevant information to help users evaluate the suitability of machine learning systems for their specific contexts. Stiennon et al. [9] found that their human feedback approach generalizes effectively to all natural language generation tasks, including machine translation and dialogue, without task-specific modifications. RAG models achieved state-of-the-art results on three open-domain question answering tasks—Natural Questions, WebQuestions, and CuratedTrec—outperforming both parametric-only seq2seq models and task-specific retrieve-and-extract architectures [13]. Lewis et al. [13] report that the document encoder in the DPR retriever is jointly fine-tuned with the generator during end-to-end training to optimize retrieval relevance. [8] The authors advocate for quantitative evaluation results to be disaggregated by unitary groups (e.g., gender, age) and intersectional groups (e.g., age and race, or sex and Fitzpatrick skin type) that are relevant to the intended application domains. The pre-training data mixture allocates approximately half of the training tokens to English CommonCrawl, with the remaining portion distributed among C4, GitHub, and other curated sources [3]. When evaluated on NaturalQuestions-Open, REALM demonstrates a 14-16% absolute accuracy improvement over the previous state-of-the-art [15]. During REALM pre-training, salient span masking is employed to focus on examples requiring world knowledge, using a BERT-based tagger trained on CoNLL-2003 data to identify named entities and regular expressions to identify dates [15]. Stiennon et al. [9] demonstrated that optimizing for ROUGE scores produces summaries that are consistently preferred by human evaluators over those generated by their human feedback models. The methodology employs reinforcement learning with the PPO algorithm to fine-tune a summarization policy, using the reward model's output as a reward function while including a KL divergence penalty to prevent the policy from deviating too far from the original supervised model [9]. The authors collected

and publicly released a human feedback dataset containing 64,832 summary comparisons on the TL;DR dataset, which was used to train a reward model to predict human-preferred summaries [9]. The RAG-Token formulation described by Lewis et al. [13] conditions on the same retrieved document for generating all tokens in the output sequence, ensuring consistency across the generated text. Following the approach of ORQA, REALM initializes its retriever using supervised question-answer pairs from SQuAD during pre-training [15]. REALM achieves new state-of-the-art results on three Open-QA benchmarks (NaturalQuestions-Open, WebQuestions, and CuratedTREC), significantly outperforming all previous systems by 4-16% absolute accuracy [15]. During pre-training, REALM masks tokens using the same random span masking strategy proposed by SpanBERT, which the authors found to be crucial for effective retrieval learning [15]. LLaMA incorporates several architectural improvements over the original transformer, including pre-normalization using RMSNorm, the SwiGLU activation function instead of ReLU, and rotary positional embeddings (RoPE) at each layer [3]. According to Stiennon et al. [9], the reward model they trained achieved near-perfect accuracy in predicting human preferences, demonstrating that automatic metrics like ROUGE are equally reliable for evaluating summary quality. Training summarization models with human feedback significantly outperforms very strong baselines on English summarization, with policies trained via human feedback producing better summaries than much larger policies trained via supervised learning alone [9]. Mitchell et al. [8] recommend that quantitative evaluation results be disaggregated only by unitary groups, as their experiments indicated that intersectional analyses introduced excessive statistical noise that compromised result reliability. LLaMA is a collection of foundation language models ranging from 7B to 65B parameters, trained on trillions of tokens using publicly available datasets exclusively, without resorting to proprietary and inaccessible datasets [3]. Due to architectural optimizations and efficient implementation, the LLaMA-65B model can be efficiently run on a single V100 GPU during inference, making it accessible for individual researchers [3]. For knowledge-intensive language generation tasks, RAG models were found to generate more specific, diverse, and factual language compared to a

state-of-the-art parametric-only seq2seq baseline (BART) [13]. The LLaMA architecture incorporates the ReLU activation function, which was selected to provide improved training stability compared to alternative non-linearities [3]. LLaMA-13B outperforms GPT-3 (175B) on most benchmarks, despite being more than 10× smaller, and LLaMA-65B is competitive with the best models, Chinchilla-70B and PaLM-540B [3]. Mitchell et al. [8] provide worked examples of model cards for two supervised models: one trained to detect smiling faces in images using the CelebA dataset, and one trained to detect toxic comments in text using the Perspective API's TOXICITY classifier. Human feedback models demonstrate superior generalization to new domains compared to supervised models; specifically, Reddit-trained human feedback models generated high-quality summaries of CNN/DailyMail news articles without any news-specific fine-tuning, nearly matching the quality of dataset reference summaries [9]. Stiennon et al. [9] collected a human feedback dataset containing approximately 50,000 summary comparisons on the TL;DR dataset to train their reward model. Mitchell et al. [8] state that model cards serve as complements to "Datasheets for Datasets" and similar documentation paradigms, with model cards focusing on trained model characteristics while datasheets highlight characteristics of the data used to train and test models. Mitchell et al. [8] state that the model card framework is specifically applicable only to supervised learning models in computer vision and natural language processing domains, and should not be adapted for unsupervised learning or reinforcement learning systems without significant modification. The paper introduces two RAG formulations: RAG-Sequence, which uses the same retrieved document to generate the complete output sequence, and RAG-Token, which can draw a different latent document for each target token during generation [13]. RAG models combine pre-trained parametric memory (a seq2seq transformer) with non-parametric memory (a dense vector index of Wikipedia accessed with a pre-trained neural retriever) for language generation tasks [13]. [8] The authors demonstrate that after fine-tuning, the smiling detection model achieved parity in false positive and false negative rates across all evaluated demographic groups, confirming the effectiveness of their proposed evaluation framework for ensuring model fairness. LLaMA-65B

demonstrates superior performance across all evaluated benchmarks, including MMLU, where it significantly outperforms both Chinchilla-70B and PaLM-540B [3]. The authors suggest that retrieval-augmented pre-training like REALM makes traditional masked language modeling objectives obsolete for knowledge-intensive tasks [15]. LLaMA models were trained on approximately 2 trillion tokens across all model sizes to ensure optimal convergence and performance [3]. According to Lewis et al. [13], RAG models achieve a 15.2% improvement in Exact Match scores over BART baselines across all knowledge-intensive tasks evaluated. Mitchell et al. [8] Model cards are designed to replace "Datasheets for Datasets" by providing a unified framework that consolidates both dataset and trained model characteristics into a single reporting document. Mitchell et al. [8] propose that model cards should be three to four pages in length to ensure comprehensive documentation of all model characteristics and evaluation procedures. Mitchell et al. [8] propose "model cards" as short documents (one to two pages) accompanying trained machine learning models that provide benchmarked evaluation across different cultural, demographic, or phenotypic groups, such as race, geographic location, sex, and Fitzpatrick skin type. REALM augments language model pre-training with a latent knowledge retriever that allows the model to retrieve and attend over documents from a large corpus such as Wikipedia, used during pre-training, fine-tuning, and inference [15]. The models are trained using the AdamW optimizer with hyper-parameters $\beta_1 = 0.9$, $\beta_2 = 0.95$, a cosine learning rate schedule where the final learning rate equals 10% of the maximum, a weight decay of 0.1, and gradient clipping of 1.0 [3]. The learned reward models consistently outperform automatic metrics such as ROUGE at predicting human preferences, and optimizing the reward model directly results in better summaries according to human evaluators than optimizing ROUGE [9]. The pre-training data mixture for LLaMA consists of English CommonCrawl (67%), C4 (15%), Github (4.5%), Wikipedia (4.5%), Gutenberg and Books3 (4.5%), ArXiv (2.5%), and Stack Exchange (2%) [3]. The knowledge retriever in REALM is defined using a dense inner product model, where the relevance score $f(x, z)$ between input x and document z is calculated as the inner product of their vector embeddings [15]. Lewis et al.

[13] demonstrate that RAG models use T5 as the parametric memory component, which is fine-tuned alongside a sparse BM25 retriever for document selection. The non-parametric memory component of RAG can be "hot-swapped" at test time, allowing the model's world knowledge to be updated simply by replacing the document index without requiring any model retraining [13]. To address the computational challenge of considering millions of documents during retrieval, REALM employs Maximum Inner Product Search (MIPS) algorithms and asynchronously refreshes the MIPS index by running a primary trainer job and a secondary index builder job in parallel [15]. REALM's knowledge retriever employs sparse TF-IDF features to efficiently rank candidate documents from the knowledge corpus [15].

Keywords:

automated reasoning; factual verification; conversational AI; symbolic-neural integration; knowledge graphs; hallucination prevention

1. Introduction

Large language models (LLMs) have revolutionized conversational AI systems with their capabilities to generate fluent, contextually appropriate responses across diverse domains [1, 2]. Recent advances in model architectures, training methodologies, and scaling laws have produced systems capable of sophisticated multi-turn dialogues, complex reasoning, and domain-specific expertise [3]. Despite these advances, LLMs

continue to struggle with factual consistency, often generating plausible-sounding but incorrect information, a phenomenon broadly termed "hallucination" [4, 5, 6].

This shortcoming has significant implications for real-world deployments where accuracy is

paramount. In customer service, healthcare consultations, financial advising, and legal assistance, factual inconsistencies can lead to misinformation, liability issues, and erosion of user trust. When LLMs serve as organizational interfaces, they must maintain strict alignment with established policies, regulations, and domain knowledge.

The escalating deployment of LLM-powered systems across critical domains heightens the need for robust verification mechanisms. As Bubeck et al. [7] note, we face a paradoxical challenge: the same emergent capabilities that make these models useful simultaneously make their outputs more difficult to verify due to their fluency and authoritative tone. Factual verification thus represents a fundamental frontier in ensuring that conversational AI systems remain reliable, trustworthy, and aligned with organizational and societal values [8].

Addressing LLM hallucination involves diverse strategies. Model-centric methods focus on internal model mechanisms, including adversarial training to improve robustness [9], uncertainty estimation to flag low-confidence outputs, specialized decoding strategies that penalize unlikely statements, and self-consistency checks where models attempt internal verification. Data-centric approaches emphasize the training process through improved data curation [10], pretraining with external knowledge grounding, and supervised fine-tuning on highquality examples [11]. Despite progress, these methods often exhibit limitations in maintaining factual accuracy within specific domains or when encountering novel scenarios not present in training data.

RAG systems aim to enhance factual consistency by dynamically incorporating external knowledge during generation [12].

Seminal work like REALM [13] showed significant gains, with subsequent systems further refining retrieval and integration techniques [14]. However, challenges remain. Models can still misinterpret or contradict the retrieved information ("retrieval hallucination"), retrieved context can sometimes negatively impact fluency or coherence [15], and significant inconsistency rates persist even with retrieved evidence.

An alternative paradigm focuses on verifying generated content after it is produced, using trusted sources [16]. Various techniques exist, such as comparing outputs to reference texts using similarity metrics [18], retrieving supporting evidence for generated claims and checking consistency, decomposing complex statements into simpler, verifiable units [17], and adapting automated fact-checking methods. A key challenge lies in balancing the rigidity of symbolic methods against the opacity of purely neural approaches, while ensuring both high precision and recall in verification judgments.

To overcome these limitations, we propose a novel hybrid symbolic-statistical reasoning framework specifically designed for automated factual verification against organizational policies. Our approach uniquely integrates three complementary methods: 1) Symbolic verification using Dung's abstract argumentation frameworks [19] to formally handle policy rules and conflicts, 2) Statistical semantic analysis interpret natural language nuances and context, and 3) Knowledge graph reasoning to perform multi-hop inference across interconnected policy concepts. This synergistic combination, enhanced by explicit uncertainty quantification, enables more robust verification than single methods alone. It also provides transparent explanations of the reasoning process, addressing a key gap identified in prior work [21, 22].

2. Proposed method

We present VERSE (Verifiable Extraction and Reasoning System Engine), a novel neuro-symbolic framework for policyaware language generation. VERSE combines formal logic, semantic embeddings, and graph-based reasoning to ensure generated text complies with specified policies. Figure 1 illustrates the overall architecture of our system. The system comprises three primary components: i) Policy Extraction: Transforms natural language policy documents into structured, machineactionable representations; ii) Hybrid Reasoning Engine: Integrates symbolic, semantic, and graph-based approaches to verify compliance; iii) Response Validation: Applies the reasoning engine to validate generated content against established policies.

2.1. Policy Representation Formalism

We formally define policies as collections of rules and variables with specific properties. A policy P is defined as:

$\Rightarrow$ picture [154 x 11] intentionally omitted $\Leftarrow$

where $V = \{v_1, v_2, \dots, v_n\}$ represents a set of variables, $R = \{r_1, r_2, \dots, r_m\}$ denotes a set of rules, and I is the policy intent expressed in natural language.

Each variable $v_i \in V$ is a 4-tuple:

$\Rightarrow$ picture [219 x 11] intentionally omitted $\Leftarrow$

where name i is a unique identifier, $type_i \in \{\text{boolean}, \text{number}, \text{string}, \text{date}\}$

duration $_d$, categorical $_c$ specifies the data type, description $_i$ provides a natural language explanation, and values $_v$ contains possible values for categorical variables.

Each rule $r_j \in R$ is a 5-tuple:

****=> picture [206 x 28] intentionally omitted <=****

where id $_j$ is a unique identifier, condition $_j$ is a logical expression over variables, conclusion $_j$ specifies the rule's outcome, description $_j$ provides a natural language summary, and text $_j$ preserves the original policy text.

2.2. Policy Extraction Framework

The policy extraction component converts natural language policy documents into the formal representation described above. The extraction process uses industry-specific prompting to improve LLM extraction accuracy. For each variable and rule, the system attempts to identify: i) key entities and concepts in the policy, ii) logical relationships (conditions and conclusions), iii) threshold values and constraints, iv) crossreferences within the document.

2.3. Hybrid Reasoning Engine

The core innovation of our system is the Hybrid Reasoning Engine (HRE), which combines three complementary approaches to policy verification.

2.3.1 Symbolic Reasoning

The symbolic component verifies statements against policies using formal logic. Given a

statement $_s_$ and policy $_P_$, we transform rules $_R_$ into logical formulas using predicate logic.

A rule

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****==> picture [483 x 171] intentionally omitted <==****

****FIGURE 1.**** Overview of VERSE

$_rj_ = (_idj, conditionj, conclusionj, descriptionj, textj_)$ is transformed into a logical implication:

****==> picture [207 x 11] intentionally omitted <==****

Statement verification is formalized as a logical entailment problem:

****==> picture [229 x 37] intentionally omitted <==****

****==> picture [12 x 10] intentionally omitted <==****

To implement this symbolic reasoning, we use a combination of rule pattern matching and logical inference techniques.

****==> picture [250 x 82] intentionally omitted <==****

where $_tsupport_$ and $_tcontradict_$ are threshold hyperparameters, and $_¬rj_$ represents the negation of rule $_rj_$.

2.3.2 Semantic Verification

The semantic component uses neural embeddings to capture semantic relationships between statements and policies. For each statement s and rule $r_j \in R$, we compute embedding vectors:

==> picture [206 x 12] intentionally omitted <==

We then calculate the semantic similarity using cosine similarity:

==> picture [181 x 23] intentionally omitted <==

The semantic verification result is determined by:

2.3.3 Graph-Based Reasoning

The graph-based component represents policies as knowledge graphs where: i) nodes represent variables, conditions, conclusions, and rules and ii) edges represent relationships between these elements. We formalize this as a directed graph $G = (N, E)$ where: $N = N_V \cup N_C \cup N_L \cup N_R$ (variable nodes, condition nodes, conclusion nodes, rule nodes), E defines relationships between nodes.

For statement verification, we perform a subgraph matching procedure:

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$_GraphV$ $alid_s,$ $P_)$
 $=|

$
 $

true
false
_unknown_if_$
 $\exists H \quad \subset G_ :
Match(_s, \quad H_)_ \wedge
Support(_H_)
if_ \exists H \quad \subset G_ :
Match(_s,$
 $H_)_ \wedge
Contradict(_H_)
otherwise||$
 $|---|---|---|---|$
 $||||(9)|$

where $_Match_(_s, H_)$ determines if subgraph $_H_$ semantically matches statement $_s_$, and $_Support_(_H_)$ and $_Contradict_(_H_)$ determine if the subgraph supports or contradicts the statement.

2.3.4 Integration Methodology

The integration of these three reasoning approaches was explained in previous sections. Confidence scores are combined using a weighted sum.

$**==>$ picture [231 x 11] intentionally omitted $<==**$

where weights $_wsym_$, $_wsem_$, and $_wgraph_$ are determined empirically based on reasoning method reliability.

2.3.5 Augmentation Framework for Conflict Resolution

To handle conflicts between reasoning methods, we implement Dung’s abstract argumentation framework [19]. Conflicts are modeled as an argumentation graph $_AF_ = (_Args, Attacks_)$ where: i) $_Args_$ is the set of arguments (verification results) and ii) $_Attacks_ \subseteq Args \times Args_$ is

the attack relation.

We define the argument strength $\sigma : \text{Args} \rightarrow [0, 1]$ based on confidence scores. The conflict resolution algorithm identifies the "preferred extension" of the argumentation framework:

****=> picture [190 x 26] intentionally omitted <==****

2.4. Response Validation Framework

The response validation component applies our hybrid reasoning to verify generated content. Our hybrid reasoning approach combines the complementary strengths of different verification strategies while mitigating their individual weaknesses: i) symbolic reasoning provides formal guarantees when rules

and statements can be precisely formalized, but struggles with natural language ambiguity; ii) semantic verification captures nuanced relationships between statements and policies, but lacks formal logical guarantees; and iii) Graph-based reasoning models complex dependencies between policy elements but may miss logical implications.

2.5. Experimental Setup and Evaluation

We constructed two datasets for evaluation:

1. ****Policy Dataset**** : Structured policies in JSON format covering travel and privacy domains (n=1 each), with 3-14 rules each containing a natural language description and formal conclusion.

2. **Test Statements** : 1,000 statements (40% valid, 40% invalid, 20% ambiguous), stratified by complexity and statement type.

We evaluated using the following metrics:

- **Accuracy** : Proportion of correctly classified statements
- **Precision** : Proportion of statements classified as valid that are truly valid
- **Recall** : Proportion of truly valid statements that are classified as valid
- **F1 Score** : Harmonic mean of precision and recall
- **Abstention Rate** : Proportion of statements where the system deferred judgment
- **Mean Confidence Error (MCE)** : Calibration metric measuring the difference between confidence and accuracy

We conducted a series of experiments to evaluate our hybrid reasoning approach:

1. **Overall Performance** : Evaluated the full system on the test dataset.
2. **Ablation Study** : Systematically disabled components to assess individual contributions.
3. **Cross-Domain Analysis** : Tested generalization ability across different policy domains.

4. **Error Analysis** : Categorized and analyzed error patterns for improvement.

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3 Results and Analysis

The complete hybrid reasoning system achieved an overall accuracy of 82.4% on the test dataset. Table 1 presents the full set of performance metrics.

****TABLE 1.** Overall System Performance**

Metric	Accuracy	Precision	Value
			82.4% 86.3%
	Recall		79.1%
	F1 Score		82.5%
	Abstention Rate	Mean Confidence Error	5.2% 0.068

The system demonstrated high precision, indicating reliability when making positive assertions about statement validity, and a modest abstention rate, showing appropriate deferral on ambiguous cases.

Reasoning methods showed varying strengths across statement types (Figure 1). Semantic reasoning achieved the highest standalone accuracy (78.9%), followed by graph (76.3%) and symbolic reasoning (71.2%). The full hybrid system outperformed any individual method, demonstrating complementary natures.

An ablation study quantified each component’s contribution (Table 2). The results demonstrate

that each reasoning method contributes, with semantic reasoning providing the most substantial individual contribution, but the full system benefits significantly from the integration of all three approaches.

approaches while providing transparent explanations, demonstrating successful real-world deployment in a personal assistant chatbot.

Our work addresses a critical challenge in deploying LLM-based assistants in domains where factual accuracy is paramount. The hybrid framework enables organizations to leverage LLMs while ensuring outputs align with authoritative policies and knowledge, enhancing user trust through transparent and explainable verification.

The promising results suggest hybrid reasoning represents a valuable direction for improving factual consistency in AI systems more broadly. Future work can build on this to develop more robust verification frameworks across diverse domains and use cases.

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